SOCI 502 Data Story

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MILESTONE 1: QUESTION AND SIMULATION

Research Question:

Do Canadian men and women choose different commuting modes and if so, does this gender difference vary by province/territory?

Dataset:

The dataset that I have selected is Table 98-10-0458-01 from the 2021 Canadian Census. This dataset includes information on Canadians' choice of commuting mode, duration of commute, time leaving for work, age, and gender, categorized by province/territory.

Plans for Data Story:

Simulated Code and Data:

```
#### Workspace Setup ####

# install.packages("tidyverse")
# install.packages("dplyr")
# install.packages("knitr")
# install.packages("tinytable")

library(tidyverse)
library(dplyr)
library(knitr)
library(tinytable)
```

```r
set.seed(723) # makes the simulation reproducible

# Naming simulated objects (note that "provinces" includes both Canadian
# provinces and territories for simplicity)

provinces <- c(
  "Newfoundland and Labrador", "Prince Edward Island", "Nova Scotia",
  "New Brunswick", "Quebec", "Ontario", "Manitoba", "Saskatchewan", "Alberta",
  "British Columbia", "Yukon", "Northwest Territories", "Nunavut"
)
```

```r
# Rough relative size of each workforce (approx. in thousands, per gender)

workforce_k <- c(90, 35, 220, 175, 2000, 3500, 290, 240, 1000, 1100, 10, 10, 10)

# Simulating provinces' workforce numbers by gender with ±10% difference

sim_grid <- expand_grid(geography = provinces, gender = c("Men+", "Women+")) |>
  mutate(
    total = round(rep(workforce_k, each = 2) * 1000 * runif(26, 0.9, 1.1))
  )

sim_grid
```

```
# A tibble: 26 x 3
   geography                  gender   total
   <chr>                      <chr>    <dbl>
 1 Newfoundland and Labrador  Men+     89529
 2 Newfoundland and Labrador  Women+   96500
 3 Prince Edward Island       Men+     35118
 4 Prince Edward Island       Women+   37654
 5 Nova Scotia                Men+    208050
 6 Nova Scotia                Women+  234297
 7 New Brunswick              Men+    167002
 8 New Brunswick              Women+  189127
 9 Quebec                     Men+   1815278
10 Quebec                     Women+ 1921510
# i 16 more rows
```

```r
# Baseline probability of using transit, set in three simple tiers (based on
# expectation, not estimates from data)
```

```r
# Large transit systems (Quebec, Ontario, BC): 8%
# Medium transit systems (Nova Scotia, Manitoba, Alberta): 4%
# All other provinces and territories: 2%

large_tier <- c("Quebec", "Ontario", "British Columbia")
medium_tier <- c("Nova Scotia", "Manitoba", "Alberta")

base_p <- case_when(
  provinces %in% large_tier  ~ 0.08,
  provinces %in% medium_tier ~ 0.04,
  TRUE                       ~ 0.02
)

base_logit <- qlogis(base_p) # convert to the log-odds scale used by the model
```

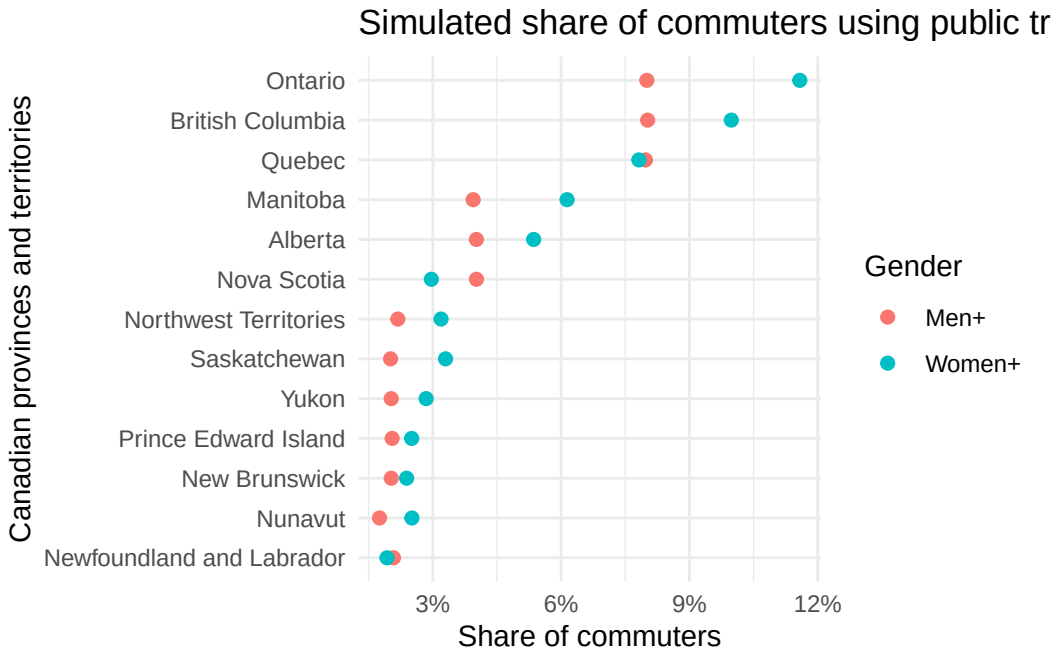
```r
# Gender effect on the log-odds scale: women are assumed to be higher on average
# (+0.25), and the size of the gap varies randomly by province.
# This variation is the interaction I want to detect.

gender_effect <- 0.25 + rnorm(13, mean = 0, sd = 0.15)

sim_grid <- sim_grid |>
  mutate(
    province_id = match(geography, provinces),
    logit_p = base_logit[province_id] +
      if_else(gender == "Women+", gender_effect[province_id], 0),
    p_transit = plogis(logit_p),
    transit = rbinom(n(), size = total, prob = p_transit),
    share = transit / total
    ) |>
  select(geography, gender, transit, total, share)

print(sim_grid)
```

```
# A tibble: 26 x 5
   geography                 gender transit   total  share
   <chr>                     <chr>    <int>   <dbl>  <dbl>
 1 Newfoundland and Labrador Men+      1863   89529 0.0208
 2 Newfoundland and Labrador Women+    1862   96500 0.0193
 3 Prince Edward Island      Men+       720   35118 0.0205
 4 Prince Edward Island      Women+     944   37654 0.0251
```

```
 5 Nova Scotia                Men+       8366  208050 0.0402
 6 Nova Scotia                Women+     6949  234297 0.0297
 7 New Brunswick              Men+       3388  167002 0.0203
 8 New Brunswick              Women+     4517  189127 0.0239
 9 Quebec                     Men+     144714 1815278 0.0797
10 Quebec                     Women+   150227 1921510 0.0782
# i 16 more rows
```

```
sim_grid |>
  tt() |>
  style_tt(j= 1:5, align = "llrrr")|>
  format_tt(digits = 4, num_mark_big = ",", num_fmt = "decimal") |>
  setNames(c("Province", "Gender", "Transit Users", "Total Commuters", "Share"))
```

```
ggplot(sim_grid, aes(x = share, y = fct_reorder(geography, share),
                     colour = gender)) +
  geom_point (size = 2) +
  scale_x_continuous(labels = scales::percent) +
  labs(title = "Simulated share of commuters using public transit",
       x = "Share of commuters", y = "Canadian provinces and territories",
       colour = "Gender") +
  theme_minimal()
```



Province	Gender	Transit Users	Total Commuters	Share
Newfoundland and Labrador	Men+	1,863	89,529	0.0208
Newfoundland and Labrador	Women+	1,862	96,500	0.0193
Prince Edward Island	Men+	720	35,118	0.0205
Prince Edward Island	Women+	944	37,654	0.0251
Nova Scotia	Men+	8,366	208,050	0.0402
Nova Scotia	Women+	6,949	234,297	0.0297
New Brunswick	Men+	3,388	167,002	0.0203
New Brunswick	Women+	4,517	189,127	0.0239
Quebec	Men+	144,714	1,815,278	0.0797
Quebec	Women+	150,227	1,921,510	0.0782
Ontario	Men+	281,395	3,515,798	0.08
Ontario	Women+	368,863	3,185,667	0.1158
Manitoba	Men+	11,062	280,457	0.0394
Manitoba	Women+	18,794	306,114	0.0614
Saskatchewan	Men+	4,740	235,507	0.0201
Saskatchewan	Women+	8,281	251,087	0.033
Alberta	Men+	40,169	999,120	0.0402
Alberta	Women+	54,140	1,010,293	0.0536
British Columbia	Men+	82,627	1,030,103	0.0802
British Columbia	Women+	119,381	1,196,369	0.0998
Yukon	Men+	187	9,218	0.0203
Yukon	Women+	285	10,018	0.0284
Northwest Territories	Men+	209	9,572	0.0218
Northwest Territories	Women+	336	10,523	0.0319
Nunavut	Men+	182	10,369	0.0176
Nunavut	Women+	270	10,750	0.0251

What Pattern Would Count as an Answer?